

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Akshaya Kumar Chandra Sekar	Reg. No.:	192425350
Section / Batch:	B. TECH AIML	Date:	23/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Divide and Conquer Algorithm Comparison with Different Combine Costs

Compare two algorithms with recurrences $T(n)=2T(n/2)+n$ and $T(n)=2T(n/2)+n^2$. Clearly define both recurrences. Apply Master Theorem to determine complexity. Use diagrams to compare structures. Interpret which is more efficient and justify.

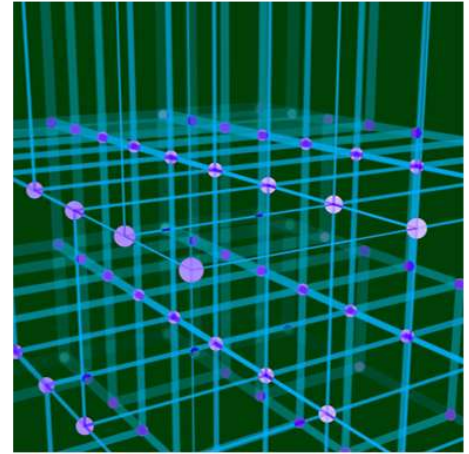


Divide and Conquer Algorithm Comparison with Different Combine Cost

Presented by:

AKSHAYA KUMAR CHANDRA SEKAR

Reg. No: 192425350



This presentation compares two divide-and-conquer algorithms with different combine costs:

- **Algorithm 1:** $T(n) = 2T(n/2) + n$
- **Algorithm 2:** $T(n) = 2T(n/2) + n^2$
- The Master Theorem is applied to determine the time complexity of both algorithms. Their recursion tree structures are compared to understand how the combine cost affects overall performance and to identify the more efficient algorithm.

What is the Master Theorem?

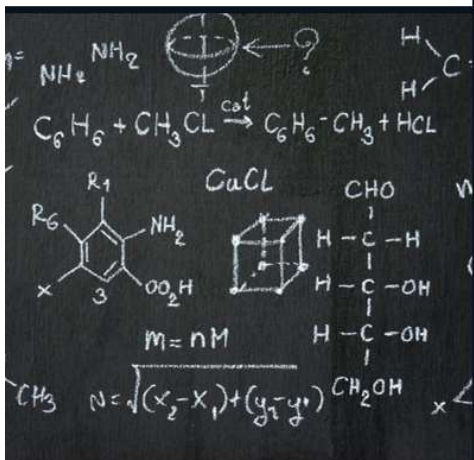
- The Master Theorem is a mathematical method used to analyze the time complexity of divide-and-conquer algorithms. It solves recurrence relations of the form:

- $T(n) = aT(n/b) + f(n)$

Where:

- a = Number of subproblems
- b = Size reduction factor
- $f(n)$ = Work done outside the recursive calls (combine cost)
- The theorem consists of three cases that help determine the overall time complexity without solving the recurrence manually. It is widely used to analyze algorithms such as Merge Sort, Quick Sort, and Binary Search.

Applications



- **Algorithm 1:** $T(n) = 2T(n/2) + n$ ($\Theta(n \log n)$)
- Merge Sort for efficient sorting.
- Large database and file sorting.
- External sorting for big data.
- Data processing in cloud computing.
- **Algorithm 2:** $T(n) = 2T(n/2) + n^2$ ($\Theta(n^2)$)
- Image and video processing with expensive merging operations.
- Scientific computing involving large matrix operations.
- Computer graphics where combining results is computationally intensive.
- Research algorithms with complex data aggregation.

Divide and Conquer Algorithm Comparison with Different Combine Costs

Comparison Problem

Compare two divide-and-conquer algorithms with recurrences $T(n) = 2T(n/2) + n$ and $T(n) = 2T(n/b) + n^2$. Apply Master Theorem to determine their time complexity. Use diagrams to compare structures. Interpret which is more efficient and Justify.

Algorithm 1

Recurrence: $T(n) = 2T(n/2) + n$

- $a = 2$ (two subproblems)
- $b = 2$ (problem size reduced by half)
- $f(n) = n$ (combine cost is linear)

Apply Master Theorem

Calculate $n^{\log_b a} = n^{\log_2 2} = n^1 = n$

Here, $f(n) = n = \Theta(n^{\log_b a})$

This matches Case 2.

Time Complexity: $T(n) = \Theta(n \log n)$

Algorithm 2

Recurrence: $T(n) = 2T(n/2) + n^2$

- $a = 2$ (two subproblems)
- $b = 2$ (problem size reduced by half)
- $f(n) = n^2$ (combine cost is quadratic)

Apply Master Theorem

Calculate $n^{\log_b a} = n^{\log_2 2} = n^1 = n$

Here, $f(n) = n^2 = \Omega(n^{\log_b a + \epsilon})$ (where $\epsilon = 1$)

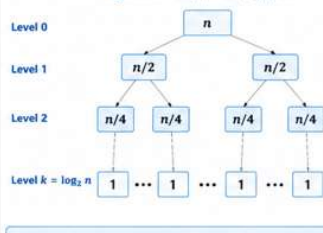
Also, $af(n/b) = 2\left(\frac{n}{2}\right)^2 = \frac{n^2}{2} \leq cn^2$ ($c = 1/2 < 1$)

This matches Case 3.

Time Complexity: $T(n) = \Theta(n^2)$

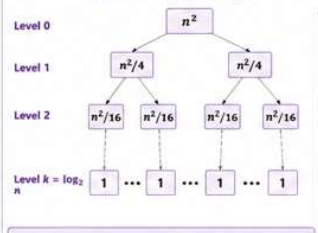
Recursion Tree Comparison

Algorithm 1: $T(n) = 2T(n/2) + n$



Each level has total cost n . There are $\log_2 n$ levels.
So, overall complexity is $O(n \log n)$.

Algorithm 2: $T(n) = 2T(n/2) + n^2$



Cost decreases geometrically, but the top level dominates.
 Complexity is $O(n^2)$.

What is Master Theorem?

Master Theorem helps to find the time complexity of recurrences of the form $T(n) = aT(n/b) + f(n)$.

Where, α = number of subproblems

b = size reduction factor

$f(n)$ = work done outside recursion (combine cost)

Three Cases of Master Theorem

Case	Condition	Time Complexity
1	If $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$	$T(n) = \Theta(n^{\log_b a - \epsilon})$
2	If $f(n) = \Theta(n^{\log_b a} \log^k n)$ for $k \geq 0$	$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$
3	If $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$ and $af(n/b) \leq cf(n)$ for some $c < 1$ and large n	$T(n) = \Theta(f(n))$

Comparison Summary

Feature	Algorithm 1 $2T(n/2) + n$	Algorithm 2 $2T(n/2) + n^2$
a (subproblems)	2	2
b (reduction factor)	2	2
f(n) (combine cost)	n	n^2
Master Theorem Case	Case 2	Case 3
Time Complexity	$\Theta(n \log n)$	$\Theta(n^2)$
Growth Rate	Lower	Higher
Efficiency	More Efficient	Less Efficient

Conclusion

- Both algorithms divide the problem into two equal subproblems.
- The main difference is the combine cost after solving the subproblems.
- Algorithm 1:
 $T(n) = 2T(n/2) + n$
 $T(n) = 2T(n/2) + n$
 $T(n) = 2T(n/2) + n$
 $T(n)$ has $\Theta(n \log n)$ time complexity.
- Algorithm 2:
 $T(n) = 2T(n/2) + n^2$
 $T(n) = 2T(n/2) + n^2$
 $T(n) = 2T(n/2) + n^2$
 $T(n)$ has $\Theta(n^2)$ time complexity.
- Therefore, Algorithm 1 is more efficient because its linear combine cost results in faster execution for large input sizes.

Code of Conduct

/ Akshaya Kumar Chandra Sekar(192425350)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya kumar c

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

